DEPARTMENT OF APEX INSTITUTE OF TECHNOLOGY

PROJECTPROPOSAL

1. Project Title: - Carbon-Aware Temporal Shifting : An Intellegent Scheduling Framework

2. Project Scope: -

Carbon-Aware Temporal Shifting: An Intelligent Scheduling Framework

The scope of this project is to design, implement, and evaluate an intelligent scheduling framework that reduces carbon emissions from computational workloads by leveraging temporal variations in carbon intensity of electricity generation. The project focuses on carbon-aware temporal shifting, where non-time-critical tasks are scheduled during periods of lower carbon intensity without violating predefined deadlines or performance constraints.

The proposed system operates in a software-based, simulation-driven environment and utilizes publicly available carbon intensity datasets or APIs to model real-world electricity emission patterns. Carbon intensity values, measured in grams of CO_2 per kilowatt-hour (gCO_2/kWh), are analyzed over discrete time intervals (hourly or sub-hourly) to identify low-emission execution windows.

The project scope includes identifying and modeling deferrable computational workloads, such as batch processing jobs, scheduled data analytics, or background computing tasks. These workloads are assumed to have flexible execution windows and clearly defined deadlines. Real-time, safety-critical, or latency-sensitive applications are explicitly excluded to ensure feasibility at the undergraduate level.

An intelligent scheduling algorithm will be developed to reschedule tasks dynamically based on predicted or historical carbon intensity data. The scheduler aims to minimize estimated carbon emissions while maintaining task completion guarantees and respecting user-defined constraints such as deadlines and task priorities. Energy consumption per task will be estimated or normalized for comparison purposes.

The framework will be evaluated through comparative simulation, where traditional time-agnostic scheduling is compared against carbon-aware temporal scheduling. Performance metrics within scope include percentage reduction in carbon emissions, task success rate, scheduling delay, and overall system efficiency. Visualization techniques such as timelines, emission graphs, and comparative charts will be used to analyze results.

The project is limited to small-scale workloads and does not involve deployment on live cloud platforms or direct interaction with physical power grids. Factors such as hardware failures, network latency, and dynamic pricing mechanisms are assumed to be ideal or constant. The framework serves as a conceptual and experimental model rather than a production-ready system.

Overall, this project demonstrates the application of intelligent scheduling and green computing principles to address sustainability challenges in computing systems, providing a practical and academically suitable solution for a Bachelor of Engineering major project.

3. Requirements: -

- Hardware Requirements
  1. Processor: Intel i5 / AMD Ryzen 5 or equivalent (minimum i3 acceptable)
  2. RAM: Minimum 8 GB (4 GB workable for basic simulations)
  3. Storage: At least 256 GB HDD/SSD
  4. Architecture: 64-bit system
  5. Internet Connectivity
     - Required for accessing:
       I. Public carbon-intensity datasets/APIs
       II. Research papers and documentation
     - No continuous high-speed connection required
- Software Requirements
  1. **Operating System**
     - Windows 10 / 11
     - OR Linux (Ubuntu 20.04 or later recommended)
  2. **Programming Language**
     - **Python 3.x**
     - Java (for object-oriented scheduling logic)
  3. **Development Tools / IDE**
     - Visual Studio Code
     - PyCharm
     - Jupyter Notebook (for analysis and visualization)

4. Libraries / Frameworks

- **NumPy** – numerical computations
- **Pandas** – dataset handling and time-series analysis
- **Matplotlib / Seaborn** – visualization of carbon intensity and scheduling results
- **Requests** – API data fetching (if live data is used)

5. **Data Sources**
   - Publicly available **carbon intensity datasets/APIs**
   - Historical or real-time data used at hourly or sub-hourly resolution
   - Data stored in CSV or JSON format
6. **Simulation & Analysis Tools**
   - Custom scheduling simulator developed in Python
   - Timeline-based task execution modeling
   - Emission estimation based on carbon intensity values
7. **Documentation Tools**
   - MS Word / Google Docs / LaTeX (for report writing)
   - MS PowerPoint (for project presentation)

STUDENTS DETAILS

Name	UID	Signature
Piyush Sirwaiya	23BCC70053	
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APPROVAL AND AUTHORITY TO PROCEED

We approve the project as described above, and authorize the team to proceed.

Name	Title	Signature (With Date)
Dr. Amardeep Singh	Carbon-Aware Temporal Shifting	